Rajas Biliye

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Education

University of Wisconsin-Madison — Madison, WI
Master of Science in Industrial Engineering (GPA: 3.80/4.0) — *Sep. 2024 – May 2026*
Relevant Coursework: Mixed Integer Non-Linear Optimization, Machine Learning, Dynamic Programming and Reinforcement Learning

Narsee Monjee Institute of Management Studies — Mumbai, India
Bachelor of Technology in Mechanical Engineering (GPA: 3.63/4.0) — *Aug. 2020 – May 2024*

Projects

Airline Crew Scheduling Optimization | *Julia, JuMP, Gurobi, MILP*
- Developed a large-scale Mixed Integer Linear Programming model in Julia/JuMP to optimize crew pairing across 150 flights and 12 airports while minimizing operating costs and satisfying FAA regulations.
- Designed a heuristic using Gurobi's Solution Pool that reduced master problem re-solves by 96%: decreasing solution time from hours to seconds.

Customer Demand & Routing Optimization | *Python, Dynamic Programming*
- Designed a simulation-based optimization model to guide routing and inventory replenishment decisions under uncertain demand, effectively balancing immediate revenue generation with long-term customer retention.
- Minimized short-sighted distribution errors by implementing rollout-based approximations, consistently outperforming baseline simulation methods in both long-term profitability and system stability.

Multi-Commodity Network Flow | *GAMS, MIP, Stochastic Programming*
- Designed a multi-commodity network flow MIP in GAMS to minimize distribution costs across geographically constrained supply chains while balancing product requirements and operational constraints.
- Improved delivery under demand uncertainty by implementing a stochastic programming model with scenario analysis in GAMS.

Experience

University of Wisconsin-Madison — Madison, WI
Grading Assistant (Advanced Optimization, Stochastic Modeling, & Logistics Modeling) — *Jan. 2026 – May 2026*
- Provided detailed written feedback on assignment submissions and identified specific errors in constraint formulation and model structure to help students understand and correct their reasoning.

Sub-Zero Group, Inc. — Madison, WI
Process Improvement Consultant — *Sep. 2024 – Dec. 2024*
- Led a manufacturing process improvement project using the Six Sigma DMAIC (Define, Measure, Analyze, Improve, Control) framework to identify production bottlenecks and 1,963 hours of annual non-value-added labor.
- Applied Lean Manufacturing principles, including 5S and facility layout optimization, to redesign assembly workflows generating over $14,000 in annual savings.

Larsen & Toubro — Mumbai, India
Engineering Intern (Defense Division) — *Jan. 2024 – May 2024*
- Led continuous process improvement initiatives on the manufacturing floor, optimizing equipment utilization and updating standard operating procedures to increase overall production throughput.

Engineering Intern (Heavy Engineering Division) — *Apr. 2023 – Jul. 2023*
- Engineered a Tableau-based vendor performance matrix to track and score 85+ global suppliers, enabling the procurement team to isolate and mitigate risk.
- Collaborated with cross-functional teams including procurement, production, and planning to support day-to-day supply chain operations using SAP IBP.

Technical Skills

Operations Research: Linear & Mixed Integer Programming, Stochastic Programming, Dynamic Programming
Languages & Libraries: Python, Julia, SQL, Scikit-Learn, GAMS, Gurobi, JuMP